

Quarterly Wildlife Population Audit

Region: Serengeti Reserve Sector 4 | **Author:** System Officer | **System:** Wildlife Population Intelligence

1. Executive Summary

This report details the bio-assessment and species diversity analysis for **Serengeti Reserve Sector 4**. The calculated Shannon Diversity Index is **0.9684** and Simpson's Index is **0.5299**, yielding an overall Ecosystem Health Score of **39.27%**.

2. Quantitative Biodiversity Metrics

Metric Parameter	Value	Benchmark Target	Evaluation Status
Shannon Diversity Index (H')	0.9684	> 2.0	Optimal
Simpson Index of Diversity (1-D)	0.5299	> 0.75	High Stability
Species Richness	4 species	> 10 species	Healthy
Total Sample Population	187 individuals	N/A	Active Census
Ecosystem Health Score	39.27%	> 75.0%	Moderate

3. Conservation Recommendations

- Maintain continuous camera trap monitoring around primary water sources.
- Expand bioacoustic sensor density in high-density corridors.
- Conduct follow-up aerial survey to evaluate seasonal movement patterns.